Investments
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Hints for Problem Set #3 The Capital Asset Pricing Model

Hints for Problem 1: To calculate the R^2, recall from regressions that:

$$R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}}$$

In this context we are considering the regression:

$$\tilde{r}^e_{i,t} = \alpha_i + \beta_i \tilde{r}^e_{m,t} + \tilde{\epsilon}_{i,t}$$

where $\tilde{r}^e_{i,t} = \tilde{r}_{i,t} - r_{f,t}$ and $\tilde{r}^e_{m,t} = \tilde{r}_{m,t} - r_{f,t}$, and:

$$\beta_i = \frac{cov(\tilde{r}^e_{i,t}, \tilde{r}^e_{m,t})}{\sigma^2_m}$$

The total variation or variance of the return i:

$$\sigma^2_i = \beta^2_i \sigma^2_m + \sigma^2_\epsilon$$

where σ_ϵ is the standard deviation of the "residual" $\tilde{\epsilon}_{i,t}$. In statistics we label the components of total variance as:

$$\underbrace{\sigma^2_i}_{\text{Total Variation}} = \underbrace{\beta^2_i \sigma^2_m}_{\text{Explained Variation}} + \underbrace{\sigma^2_\epsilon}_{\text{Unexplained Variation}}$$

In finance we label these as:

$$\underbrace{\sigma^2_i}_{\text{Total Variance}} = \underbrace{\beta^2_i \sigma^2_m}_{\text{Systematic Risk}} + \underbrace{\sigma^2_\epsilon}_{\text{Idiosyncratic Risk}}$$

The R^2 of the regression for security i is:

$$R^2_i = \frac{\beta^2_i \sigma^2_m}{\sigma^2_i}$$